

SMART CLINIC DATABASE SYSTEM

Mid-Project Progress Report

Stage 1: Database Design and Initial Implementation

Reporting Point: End of Stage 1 - before Task 3

Table 1. Team Members and Primary Contributions

Student Name	Student ID	Primary Contribution
Sultan Aljohani	240035689	Group Leader and Repository Manager, Task 1: Database Design
Amjad Najmi	240020509	Task 2: Database Implementation
Abdulaziz Alharbi	250021379	Task 3: SQL Operations

1. Progress Summary

At this reporting point, the project has reached the end of Stage 1. The team has defined the system scope, allocated responsibilities, completed the database design, and prepared the core MySQL implementation for validation. The design maps ten related tables and includes the required EER specialization, integrity constraints, and fictional Saudi-context data. Full MySQL Workbench verification, Task 2 screenshot evidence, Task 3 SQL operations, and the Project Reflection remain scheduled for the next stage.

2. Midpoint Progress Status

Table 2. Stage 1 Workstream Status

Workstream	Midpoint Status	Evidence / Expected Output
Project planning and responsibility allocation	Completed	Contribution record, approved repository structure, and delivery plan.
Task 1 Database Design	Completed and reviewed	Ten-relation EER model, relational schema, keys, cardinalities, assumptions, and business rules.
Task 2 Core Implementation	Prepared for validation	MySQL database setup, ten CREATE TABLE statements, constraints, and fictional sample data.
Task 2 Workbench Evidence	Planned for next stage	Successful execution, populated-table results, and readable screenshots.
Task 3 SQL Operations	Planned for next stage	SELECT, JOIN, nested query, GROUP BY, UPDATE, DELETE, VIEW, and TRIGGER.
Task 4 Project Reflection	Planned for next stage	A 300-500-word reflection written after all technical work and evidence are complete.

3. Work Completed at the Midpoint

- ✓ Defined ten connected relations: CLINIC_PERSON, PATIENT, SPECIALTY, DOCTOR, APPOINTMENT, CONSULTATION, MEDICATION, PRESCRIPTION, PRESCRIPTION_ITEM, and PAYMENT.
- ✓ Modeled CLINIC_PERSON as the supertype and PATIENT and DOCTOR as total, disjoint subtypes.
- ✓ Specified primary keys, foreign keys, unique keys, CHECK constraints, and controlled ENUM values.
- ✓ Mapped relationships and optional cardinalities between appointments, consultations, prescriptions, medication items, and payments.
- ✓ Prepared a repeatable MySQL 8.0 script using utf8mb4 and a dependency-aware table creation and insertion order.
- ✓ Conducted an initial consistency review between the EER design, relational schema, and core SQL implementation.

4. Key Design Decisions

Table 3. Design Decisions and Rationale

Decision	Reason
Shared CLINIC_PERSON supertype	Avoids repeating identity and contact attributes in separate patient and doctor tables.
Total, disjoint specialization	Gives each clinic person one clear subtype while demonstrating the required EER feature.
Separate SPECIALTY relation	Normalizes specialty details and lets multiple doctors share one controlled specialty record.
Separate APPOINTMENT and CONSULTATION	Distinguishes planned visits from clinical outcomes and allows an appointment to exist before consultation.
Associative PRESCRIPTION_ITEM relation	Resolves the prescription-medication many-to-many relationship while storing dosage, route, duration, and quantity.
Multiple PAYMENT records per appointment	Supports installments, insurance processing, refunds, and different payment statuses.
One evolving SQL file	Keeps the design, implementation, later operations, VIEW, and TRIGGER in one controlled project source.

5. Challenges and Responses

Table 4. Stage 1 Challenges and Responses

Stage 1 Challenge	Response
Aligning the EER model with the relational schema	Cross-checked every entity, key, relationship, and cardinality against the SQL design before implementation testing.
Maintaining referential insertion order	Ordered parent tables before dependent clinical and financial tables so every foreign key can be satisfied.
Preventing schedule conflicts	Added unique doctor-time and patient-time constraints to APPOINTMENT.
Representing optional clinical events	Separated APPOINTMENT, CONSULTATION, and PRESCRIPTION so records are created only when each real event occurs.
Maintaining realistic and valid sample data	Used linked Saudi-context records, unique identifiers, +966 validation rules, and CHECK constraints.

6. Team Contributions at the Midpoint

Table 5. Recorded Stage 1 Contributions

Member	Midpoint Contribution	Status	Output
Sultan Aljohani	Team coordination, repository planning, requirements, entities, keys, relationships, cardinalities, EER specialization, and schema alignment.	Completed and reviewed	Task 1 design package and repository organization.
Amjad Najmi	Database creation, ten table definitions, data types, constraints, insertion order, and fictional sample records.	Prepared for validation	Core smart_clinic_database.sql implementation.
Abdulaziz Alharbi	Reviewed Task 3 requirements and prepared the operation, testing, and screenshot checklist for the next stage.	Planned for next stage	Task 3 work and evidence plan.
All members	Cross-review of Stage 1, design-implementation consistency checking, milestone planning, and preparation of this progress report.	In progress	Review notes and Mid-Project Progress Report.

7. Group's Plan for Next-Stage.

1. Review and approve the complete Stage 1 EER model, relational schema, assumptions, constraints, and sample data as a team.
2. Run the complete Stage 1 script in MySQL Workbench and correct any environment-specific execution issues.
3. Capture and organize the required Task 2 screenshots showing the populated tables and successful execution.
4. Implement and test Task 3 SELECT, JOIN, nested, GROUP BY, UPDATE, DELETE, VIEW, and TRIGGER operations in the same SQL file.
5. Capture Task 3 code-and-result evidence and connect each image to the Screenshot Evidence Index.
6. Write the Project Reflection after the technical work, results, and team contributions have been verified.
7. Complete the README and final report, then verify all file names, links, Word/PDF outputs, and submission requirements.